Authored by : Arpit Raj , Lnmiit jaipur

DATABASE NORMALIZATION NOTES

A relation is in First Normal Form (1NF) if:

- Every attribute contains atomic (indivisible) values.
- Every row is unique.
- There are no repeating groups.
- Every column stores values of a single type.

Repeating Groups

Definition

A repeating group occurs when the same type of information is stored multiple times in one row.

Example

StudentID	Phone1	Phone2	Phone3
101	9876	9123	8456.

Why are Repeating Groups Bad?

Suppose Aadya buys five books.

The table only has:

- Book1
- Book2
- Book3

Where will Books 4 and 5 go?

The design is inflexible.

Convert to 1NF.

StudentID	Skills
101	C++,Python

Solution

StudentID	Skill
101	C++
101	Python,

Repeated rows are acceptable.

Only multiple values inside one cell violate 1NF.

Second Normal Form (2NF)

A relation is in Second Normal Form (2NF) if:

- It is already in 1NF.
- Every non-prime attribute is fully functionally dependent on the entire Candidate Key. (i.e partial dependancy must be removed)

2NF mainly applies when a table has a Composite Candidate Key.

Converting to 2NF

Original

StudentID	CourseID	StudentName	Grade
value	*value*	*value*	*value*

Split into

Student

StudentID	StudentName
value	*value*

Enrollment

StudentID	CourseID	Grade
value	*value*	*value*

> Now
>
> StudentName depends entirely on StudentID,
>
> and Grade depends entirely on (StudentID, CourseID).
>
> No Partial Dependency remains.

> a single-column Primary Key automatically satisfy 2NF